Rigid Analytic Geometry Week 2 Solution

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Problem 1

Let $e_1, \cdots, e_m$ be the elements of the standard basis. Then

$$\|e_i|K^m\| = 1, x = \sum_{i=1}^{m} x_i e_i, x_i \in K.$$

Let $C \in K$ be such that

$$C = \max\{\|e_i\|_1 \mid i = 1, \cdots, m\}.$$

Then,

$$\|x\|_1 \leq C \max\{|x_i|_K \mid i = 1, \cdots, m\} = C\|x|K^m\|.$$

Problem 2

Suppose for all $C > 0$ there is $x \in K^m$ such that

$$\|x|K^m\| > n\|x\|_1.$$

Take a sequence $(x_n)_{n\in\mathbb{N}}$ in K^m such that

$$\|x_n|K^m\| > n\|x_n\|_1.$$

Then set $(y_n)_{n\in\mathbb{N}}$ be such that

$$\forall n \in \mathbb{N}, y_n = \frac{x_n}{x_n^{n_i}},$$

where $x_n^{n_i}$ is the coefficient of x_n in the standard basis with the largest absolute value. Since $(y_n)_{\mathbb{N}}$ is a bounded sequence and K^m is Banach, there is a convergent subsequence of it with its limit y. Then, taking the limit in

$$1 > \frac{\|x\|_1}{\|x|K^m\|} n = \|y_n\|_1 n.$$

we see that

$$1 > \|y\|_1 \lim_{n\to\infty} n.$$

But $\|y\|_1 \neq 0$ since

$$1 = \lim_{n\to\infty} \|y_n|K^m\| = \|y|K^m\|.$$

Thus $\|y\|_1 \neq 0$ as $y \neq 0$. This is a contradiction.

Problem 3

Consider p and q-adic norm and $\mathbb{Q}$ where p, q are distinct primes. Then for any $n \in \mathbb{N}$, consider

$$\left|\left(\frac{1}{p}\right)^n\right|_p = p^n, \left|\left(\frac{1}{p}\right)^n\right|_q = 1.$$

Thus $|\cdot|_p, |\cdot|_q$ are not equivalent.

Problem 4

For $x \in L$, consider,

$$m_x(y) = xy.$$

Then m_x is a linear operator between Banach algebras thus bounded. Also

$$T : L \ni x \mapsto m_x \in \mathrm{Hom}(L, L)$$

is also a linear operator in finite dimensional Banach spaces thus bounded. We conclude,

$$\|m_x\|_{\mathrm{op}} \leq C\|x\|.$$

Thus,

$$\|xy\| \leq \|m_x\|\|y\| \leq C\|x\|\|y\|.$$

Problem 5

Let $x \in \mathfrak{g}_{\mathfrak{U}}$, then

$$\{\mathfrak{p} \in \mathrm{Spec}\, A \mid x \in \mathfrak{p}\} \in \mathfrak{U}.$$

Thus,

$$\{\mathfrak{p} \in \mathrm{Spec}\, A \mid x \in \mathfrak{p}\} \subseteq \{\mathfrak{p} \in \mathrm{Spec}\, A \mid ax \in \mathfrak{p}\}.$$

As $\mathfrak{U}$ is a filter, $\{\mathfrak{p} \in \mathrm{Spec}\, A \mid ax \in \mathfrak{p}\} \in \mathfrak{U}$. Thus $ax \in \mathfrak{g}_{\mathfrak{U}}$.

Problem 6

Let $X = \mathrm{Sper}\, A$. Then X is T_0 since for any distince $x, y \in X$, there exists an element a such that $a \in x$ but $b \notin y$. Thus $\mathcal{P}(-a)$ will separate them.

Let Z be an irreducible closed subset of X. Consider

$$C_Z = \bigcup_{C \in Z} C.$$

Then C_Z is clearly a proper cone. To see this is prime, note that if there is $a \in A$ such that $a \in C, -a \notin C, a \notin C', -a \in C'$ for some $C, C' \in Z$. Then we have two non-empty open subsets $\mathcal{P}(a), \mathcal{P}(-a)$ of Z whose intersection is empty. This contradicts to the fact that Z is irreducible. Thus C_Z is prime. Since $C_Z \subseteq C$ for any $C \in Z$, we conclude it is a generic point. In particular X is sober. Couldn't figure out how to show that the topology base contains only quasi-compact sets.

Problem 7

Sobriety is clear as all closed subsets are finites set of points not containing η.

Thus each irreducible closed subset admits a unique generic point which is itself. Also all open sets are quasi-compact as for any open covering (U_i), let

$$U = \bigcup_i U_i$$

then $X \backslash U$ is finite. Thus take any U_1 and as $X \backslash U_1$ is finite and contains $X \backslash U$, we only have to take at most $|X \backslash U_1| - |X \backslash U|$ many more open sets to cover U.

Problem 10

Up shot : On a complete valuation field K, given an irreducible polynomial $f \in K[x]$, then any $K[x]$ close enough to f of the same degree is automatically irreducible, giving the same field extension as f. Let L/K be a Galois extension. There is $\alpha \in L$ such that $L = K(\alpha)$. Take

$$f = x^n + a_{n-1}x^{n-1} + \cdots + a_0,$$

be ethe minimal polynomial of α. Take,

$$g(x) = x^n + b_{n-1}x^{n-1} + \cdots + b_0$$

such that $\|f - g\| < \max_{0 \leq i < n} |a_i|^{\frac{1}{n-i}}$. Take β to be a root of g. Let $\alpha_1 = \alpha, \alpha_2, \cdots, \alpha_n$ be roots of f, then

$$\left| \prod_{1 \leq i \leq n} (\beta - \alpha_i) \right| = |f(\beta)| = |f(\beta) - g(\beta)| \leq \|f - g\| \max\{|\beta|^n, 1\}.$$

Finally, by sheet 4 problem 6

$$\leq \|f - g\| \cdot \max\{|a_i|^{\frac{1}{n-i}}\}.$$

Take g close enough to f such that

$$\|f - g\| \cdot \max\{|\alpha_i|^{\frac{1}{n-i}}\} < \min_{1 \leq i < j \leq n} |\alpha_i - \alpha_j|.$$

There is $1 \leq i_0 \leq n$ such that

$$|\beta - \alpha_{i_0}| < \min_{i \neq j} |\alpha_i - \alpha_j|.$$

By Krasner's lemma,

$$\alpha_{i_0} \in K(\beta).$$

On the other hand,

$$n = [L : K] \leq [K(\beta) : K] \leq [K(\beta) : K] \leq n.$$

In particular $[K(\beta) : K] = n$, thus g is irreducible over K. Let $\beta_1 = \beta, \beta_2, \cdots, \beta_n$ be roots of g. Since L/K is Galois $\beta_1, \cdots, \beta_n \in L$¿ Let

$$L' := K(\beta_1, \cdots, \beta_n).$$

Since $K(\beta)$ is dense in L, so is L'. Also L is the splitting field of g, L/K is Galois.

Problem 11 Skipped, general topology argument.

Problem 12

Take

$$U = \mathcal{R}(f_0|f_1, \cdots, f_n).$$

Take $\varepsilon \in K^\times$ such that $|\varepsilon| < 1$. Then for any $N > 0$,

$$\langle f_0, f_1, \cdots, f_n, \varepsilon^N \rangle_{\mathbb{T}_1} = \mathbb{T}_1.$$

Moreover,

$$U = \bigcup_{N>0} \mathcal{R}(f_0|f_1, \cdots, f_n, \varepsilon^N).$$

Example 0.1. *Consider $\mathbb{T}_{1,K}$ where $K = \overline{K}$. Then,*

$$\operatorname{Sp} \mathbb{T}_1 = K^\circ.$$

Taket $\varepsilon \in K^\times$ with $|\varepsilon| < 1$ and ,

$$U_N := \mathcal{R}(\varepsilon^N|X), V := \mathcal{R}(1|X) \cap \mathcal{R}(X|1) = \mathcal{R}(X|1, X^2).$$

Then

$$U_N \Leftrightarrow \{l \in K^\circ \mid |l| < |\varepsilon_N|\}, V \Leftrightarrow \{l \in K^\circ \mid |l| = 1\}.$$

Then $\operatorname{Sp} \mathbb{T}_1 = \bigcup_{N \geq 1} U_N \cup$. But this cover is not in $\mathcal{I}_{\mathbb{T}_1}$, because it dose not have a finite subcover.

Problem 14

Let $U = \mathcal{R}(f_0|f_1, \cdots, f_n), V = \mathcal{R}(g_0|g_1, \cdots, g_m)$. We claim that

$$U \cap V = \mathcal{R}(f_0 g_0 | f_i g_j, 0 \leq i \leq n, 0 \leq j \leq m).$$

Indeed,

$$|f_0(x)|, |g_0(x)| \neq 0 \iff |(f_0 g_0)(x)| \neq 0,$$

$$\langle f_i g_j \mid 0 \leq i \leq n, 0 \leq j \leq m \rangle_A = \langle f_i \mid i \rangle_A \cdots \langle g_j \mid j \rangle_A = A.$$

$$\forall i, j, \begin{cases} |f_0(x)| \geq |f_i(x)|, \\ |g_0(x)| \geq |g_j(x)|. \end{cases} \iff \begin{cases} |f_0 g_0(x)| \geq |f_i g_0(x)|, \\ |f_0 g_0(x)| \geq |f_0 g_j(x)|. \end{cases}$$